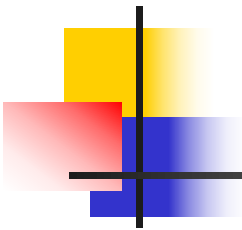
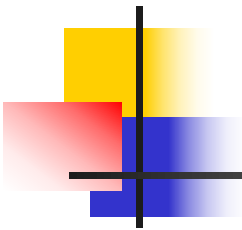


# MySQL事务

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- 
- MySQL默认在每一条SQL语句执行后都自动提交（事务），也允许在存储过程中使用显式地事务控制。
  - 事务控制原则
    - 根据需要手工启动事务
    - 根据处理情况（成功时）提交事务或（失败时）回滚事务
  - 常用的事务控制语句
    - START TRANSACTION ——用于启动事务
    - COMMIT ——用于提交事务
    - ROLLBACK ——用于回滚事务



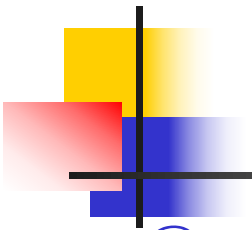
## 【例】银行账户转账

### ■ 该事务的要求

- 转账先向转入账户增加余额，再从转出账户扣减余额
- 如遇转出账户不够扣减，则放弃所有更改（即回滚事务）

### ① 创建账户关系

```
CREATE TABLE `bank_account`  
( `Account_Id` INT PRIMARY KEY,  
  `Balance` INT UNSIGNED NOT NULL )  
COMMENT='用于展示事务的银行账户表' ENGINE=INNODB;
```



## ② 定义事务

#创建过程transfer, 定义转帐事务

delimiter \$

```
CREATE PROCEDURE transfer(IN 转出账户 INT,IN 转入账户 INT,IN 转账金额 INT,OUT 转账结果 INT)
```

```
MODIFIES SQL DATA
```

```
BEGIN
```

```
    DECLARE 转出账户余额 INT;
```

```
    DECLARE exit handler FOR SQLEXCEPTION BEGIN ROLLBACK; SET 转账结果=-1;END;
```

```
    START TRANSACTION;
```

```
    UPDATE bank_account SET Balance=Balance+转账金额 WHERE Account_Id=转入账户;
```

```
    SELECT Balance INTO 转出账户余额 FROM bank_account WHERE Account_Id=转出账户;
```

```
    if 转出账户余额<转账金额 then
```

```
        ROLLBACK;
```

```
        SET 转账结果=0;
```

```
        SELECT '账户余额不足, 转账失败! ';
```

```
    ELSE
```

```
        UPDATE bank_account SET Balance=Balance-转账金额 WHERE Account_Id=转出账户;
```

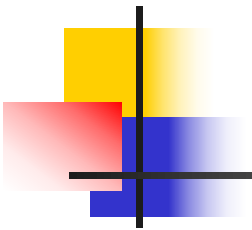
```
        COMMIT;
```

```
        SET 转账结果=1;
```

```
        SELECT '转账成功! ';
```

```
    END if;
```

```
END $
```



### ③ 执行业务

- **SET** @status\_transfer=100;
- **SET** @out\_account=101;
- **SET** @in\_account=102;
- **SET** @transfer\_balance=500;
- **SELECT \* FROM** bank\_account;
- **CALL**  
transfer(@out\_account,@in\_account,@transfer\_balance,  
@status\_transfer);
- **SELECT \* FROM** bank\_account;
- **SELECT** @status\_transfer;